



Letter to the Editor

Therapeutic hypothermia after cardiac arrest: The Mediterranean-diet Italian style

Sir,

The optimal timing of therapeutic hypothermia after out-of-hospital cardiac arrest is uncertain; current evidences suggest it should be induced within 90–120 min, resulting in a target core temperature <4 h from the event.¹ A delay in cooling the patient results in worse benefits.² Recent guidelines³ report that hypothermia need to be started during cardio-pulmonary resuscitation after the patient's heart is shocked back into beating, assuming that the earlier the cooling target is achieved, the greater is the benefit.

Actual cooling methods include the rapid intravenous ice-cold crystalloid fluid and surface cooling with ice packs or cooling blankets.⁴ A novel pre-hospital trans-nasal cooling device has been recently proposed,⁵ but it may be difficult to carry (12 kg weight) from ambulance to patient's bed, climbing narrow staircases to reach small Italian flats.

In our country over 75% of cardiac arrests occur at home. To solve the problem of a rapid hypothermia a new "Italian style" method with typical Mediterranean-diet foodstuff is becoming frequently practice, as exemplified in the following case. A 65-year-old man, developed loss of consciousness at home and went in pulse-less electric activity. An emergency code was immediately activated and an ACLS protocol initiated within 5 min. The patient returned to spontaneous circulation within 10 min from the beginning of resuscitation maneuvers. Hypothermia was induced during ACLS by putting, on body surface, ice packs and any sort of ice-frozen foodstuffs (bags of beans, cauliflowers, cherry tomatoes and anything at hand) taken from the patient's freezer. Within 6 more minutes the patient was transferred to the ambulance and a rapid infusion of large volumes of normal saline and surface cooling blankets were used for the maintenance phase of hypothermia. The patient's initial core temperature was 37.0 and vital signs included a systolic blood pressure of 90 mmHg, heart-rate of 120 beats/min and pulse oximetry (SpO₂) of 95% on 100% FiO₂.

On arrival in the Emergency department 40 min after the beginning of hypothermia, the patient was deeply comatose with absent brain stem oculocephalic reflexes, with a core temperature of 32 °C measured by a thermistore bladder catheter, 28 min after cardiac arrest. His body was covered with any sort of vegetables (see Fig. 1), mainly in areas of superficial arteries. The target temperature was maintained for 24 h with infusions of large volumes of ice-cold (4 °C) normal saline and surface cooling blankets. The patient was then slowly re-warmed by a passive method, weaned from vasopressors, sedation and paralytics, then extubated on day-3. The patient fully recovered and was discharged neurologically intact on day-12.



Fig. 1. Ice packs with frozen foodstuffs like beans, cauliflowers and cherry tomatoes taken from the patient's freezer are placed on the body surface to induce hypothermia.

While waiting for cooling kits (when available), the need to obtain early cooling encourages the EMS personnel to use any available tool, mainly frozen vegetables or any food taken from the household freezer of the patient with cardiac arrest. The strength of guideline recommendations is translated into imaginative solutions by our emergency personnel, in order to earlier obtain the target temperature. Today, no data are available to define whether Mediterranean-style foodstuff is better than Western diet choices.

Conflict of interest statement

None to declare.

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